Carmichael ∩ Lucas–Carmichael numbers (Williams numbers) with four prime factors

Prior work (added 2026-11-01): these are the *Williams numbers* of Echi (2007); McIntosh–Dipra, J. Number Theory 147 (2015) proved finiteness with d−3 primes fixed by the product-of-quotients mechanism used below and showed four-prime examples have $p_1 > 2\times10^4$, $N > 10^{24}$. **After reading their paper in full (2026-11-01): Theorem A below IS their Theorems 2–3 for d = 4** (they have $p_2 < 5p_1^{3/2}$, $p_3 < p_1p_2$, $p_4 < \sqrt{(2p_1p_2p_3)}$, and prove the quotient product exceeds $8p_1^3$ via Newton's inequality). Our proof is independent; Theorem B extends their exclusion. See `docs/literature-williams-numbers.md`.

Working note, version 2026-11-01 (first version 2026-10-29). Code, logs and result files: `clc/k4/`.

1. Statement

A squarefree $n = p_1\cdots p_k$ is **CLC** if $(p-1) \mid (n-1)$ and $(p+1) \mid (n+1)$ for every $p \mid n$. Williams (1977) showed $k \neq 3$ and $3 \nmid n$. Four prime factors is the first open case.

Theorem A. Let $n = p_1p_2p_3p_4$ $(p_1<p_2<p_3<p_4)$ be CLC. Then (a) every $p_i \equiv 1 \pmod{12}$; (b) $p_3 < 2p_1p_2$, $p_4 < \sqrt{(2p_1p_2p_3)} < 2p_1p_2$, and $n < 4(p_1p_2)^3$; (c) $p_2^2 \le 24p_1^3 + 1$. Consequently $p_3, p_4 < 9.9\,p_1^{5/2}$, $n < 477\,p_1^{15/2}$, and for each p_1 there are only finitely many such n.

Theorem B (computation). There is no 4-prime CLC number with $n \le 10^{30}$, and none whatsoever with $p_1 \le 500{,}000$. Hence any 4-prime CLC number has $p_1 > 500{,}000$ and $n > 10^{30}$.

2. Proofs

Lemma 1. For $p \mid n$: n is CLC at $p \iff n/p \equiv 1 \pmod{L_p}$, $L_p := (p^2-1)/2$. *Proof.* $(p-1)\mid(n-1)$ and $(p+1)\mid(n+1) \iff n \equiv 1 \pmod{p-1}$ and $n \equiv -1 \pmod{p+1} \iff n \equiv p \pmod{L_p}$, since $L_p = \operatorname{lcm}(p-1, p+1)$; and $L_p \mid n - p = p(n/p - 1) \iff L_p \mid n/p - 1$, since $\gcd(p, p^2-1) = 1$. ∎

(a). $4 \mid L_p$ for every odd p, so $n \equiv p \pmod 4$ for all $p \mid n$: all p_i lie in one class $c \bmod 4$ and $n \equiv c^4 \equiv 1$, so $c = 1$. For $p \neq 3$, $3 \mid L_p$, so $n \equiv p \pmod 3$ for every $p \mid n$ with $p \neq 3$; if $3 \mid n$ this forces every other prime $\equiv 0 \pmod 3$, absurd, so $3 \nmid n$ (this re-proves Williams's observation) and all p_i lie in one class $c' \bmod 3$ with $n \equiv c'^4 \equiv 1 \pmod 3$, so $c' = 1$. ∎

(b). Write $P = p_1p_2$, $q = p_3$, $r = p_4$. Lemma 1 at r: $Pq - 1 = d\cdot L_r$ with an integer $d \ge 1$, i.e. $2Pq = d(r^2-1) + 2$. (iii) Hence $2Pq \ge r^2 + 1 > q^2$, so $q < 2P$ and $r < \sqrt{(2Pq)} < 2P$; thus $m := qr < 4P^2$ and $n = Pm < 4P^3$. Also $d < 2Pq/(r^2-1) < 2P/q < 2p_1$. Likewise Lemma 1 at q and at p_2 give integers $e, s \ge 1$ with $2Pr = e(q^2-1) + 2$, $2p_1qr = s(p_2^2-1) + 2$. (iv), (ii) (Equivalently $2n = t\cdot p^3 - (t-2)\cdot p$ for every $p \mid n$, with $t = (n/p - 1)/L_p$.) ∎

(c). Multiply (iii), (iv), (ii): the left side is $8p_1^3p_2^2q^2r^2$; the right side is $d{\cdot}e{\cdot}s \cdot r^2q^2p_2^2 \cdot \Pi$, $\Pi := (1 - (d-2)/(dr^2))\,(1 - (e-2)/(eq^2))\,(1 - (s-2)/(sp_2^2))$. So $d{\cdot}e{\cdot}s{\cdot}\Pi = 8p_1^3$. Each factor of Π lies in $[1 - 1/x^2,\ 1 + 1/x^2]$ with $x \in \{r, q, p_2\} \ge p_2$, so $(1 - 1/p_2^2)^3 \le \Pi \le (1 + 1/p_2^2)^3$. Now read (iii), (iv), (ii) modulo p_1: the left sides vanish, so $d(r^2-1) \equiv e(q^2-1) \equiv s(p_2^2-1) \equiv -2 \pmod{p_1}$; as $p_1 \ge 13$ is odd, $p_1 \nmid d, e, s$. Hence $d{\cdot}e{\cdot}s \ne 8p_1^3$, i.e. $\Pi \ne 1$. If $\Pi < 1$ then $des \ge 8p_1^3 + 1$ and $\Pi = 8p_1^3/des \le 1 - 1/(8p_1^3+1)$; with $\Pi \ge 1 - 3/p_2^2$ this gives $p_2^2 \le 24p_1^3 + 3$. If $\Pi > 1$: a factor of Π exceeds 1 only when its quotient is 1; $e > d \ge 1$ (as $r > q$) and $s > 2p_1$ (as $qr \ge p_2^2+1$), so only the d-factor can, and $\Pi \le 1 + 1/r^2$. With $des \le 8p_1^3 - 1$ we get $\Pi \ge 1 + 1/(8p_1^3-1)$, hence $r^2 \le 8p_1^3 - 1$ and $p_2^2 < r^2 < 8p_1^3$. In either case $p_2^2 \le 24p_1^3 + 1$ (the first case via $p_2^2 \equiv 1 \bmod 24$). ∎ **Correction (2026-10-29):** the first write-up handled $\Pi > 1$ with "$(1+1/p_2^2)^3 \le 1 + 3.0001/p_2^2$", which is false (coefficient 3.0022 at $p_2 = 37$) and does not close; caught by the independent fact-check (docs/review-facts-2026-10-29.md). The enumeration used $p_2^2 \le 24p_1^3+3$, satisfied in both cases, so Theorem B stands.

Consequences. $p_2 < 4.9p_1^{3/2}$; $p_3 < 2P < 9.8p_1^{5/2}$; $p_4 < 2P$; $n < 4P^3 < 4{\cdot}(4.9)^3{\cdot}p_1^{15/2} < 477p_1^{15/2}$.

Check on a relaxed instance. With the modulus $(p^2-1)/24$ instead of $(p^2-1)/2$ the same algebra gives $d{\cdot}e{\cdot}s{\cdot}\Pi = 24^3p_1^3$, and the solution $13685 = 5{\cdot}7{\cdot}17{\cdot}23$ has d=27, e=67, s=977: des = 1,767,393, Π = 0.977650, $des{\cdot}\Pi = 1{,}728{,}000 = 24^3{\cdot}5^3$. ✓

3. Algorithm (`clc/k4/enum_k4.c`)

For each admissible pair $p_1 < p_2$ (both $\equiv 1 \bmod 12$; $p_2 \le \min((B/p_1)^{1/3}, \sqrt{(24p_1^3+3)})$): 1. Lemma 1 at p_1 and p_2: $p_2m \equiv 1 \pmod{L_1}$, $p_1m \equiv 1 \pmod{L_2}$. Solvable iff $p_1 \equiv p_2 \pmod{\gcd(L_1,L_2)}$ (about 23% of pairs); then m lies in one class mod $\Lambda = \mathrm{lcm}(L_1,L_2) \approx P^2/(4{\cdot}\gcd)$. 2. Candidates $m \equiv m_0 \pmod{\Lambda}$ in $(p_2^2, \min(4P^2, B/P)]$. With $j := d{\cdot}e$, (iii)·(iv) gives $jm^2 - 4P^2m = j(q^2+r^2-1) - 4P(q+r) + 4$, hence $|jm - 4P^2| \le \max(2j\sqrt{(2p_1)}, 8p_1)$. The admissible j for a fixed m form an interval containing $4P^2/m$, so testing $\lfloor 4P^2/m \rfloor$ and $\lceil 4P^2/m \rceil$ is an exact filter (the true j itself need not be either — the self-tests caught this). For large m the walk goes over j instead of m. 3. Survivors: base-2 Fermat test on n (necessary for Carmichael numbers), then r is recovered as an integer root of $dx^3 - (d-2)x - 2n$ for $d \le 2p_1$, $q = m/r$, and all four conditions plus primality are checked; everything is re-verified independently in Python (`verify_k4.py`). Cost: $\sim B^{1/2}/\log^2 B$ pairs for the n-bounded run, $\sim X^{5/2}$ for the p_1-bounded run; 44 and 10 million pairs/s in the two campaigns on the M4 Pro.

Validation. Brute-force self-tests on the (q,r) half-problem in four relaxed variants (43 half-solutions) check (iii), the window inequality, the cubic, and both reconstruction paths; the CRT class is brute-forced for 13,037 pairs; the two-sweep walk reproduces a single-sweep reference on six instances (up to 11,007,875 survivors); on the relaxed problem the enumerator finds exactly the one solution (13685) that brute force finds below 10^{12}; on the real problem `--B 1e24` reproduces the empty result of the complete 10^{24} scan in 6 s.

4. Results (2026-10-29, Apple M4 Pro, 14 threads)

scope	(p_1,p_2) pairs	window survivors	Fermat survivors	solutions	wall
$n \le 10^{30}$	230,612,582,193	1,904	1	0	87 min

scope	(p_1,p_2) pairs	window survivors	Fermat survivors	solutions	wall
$p_1 \le 500{,}000$, any n	86,517,004,140	1,185	1	0	146 min

The one Fermat survivor is the Carmichael number 26921089 = 13·37·97·577 ($37+1 \nmid n+1$). Logs: `clc/k4/k4_*.jsonl` (one line per block).

5. Heuristics and what remains

Conditioning on (a), each condition $n/p \equiv 1 \pmod{L_p/12}$ has "probability" $24/p^2$. Summing $\prod 24/p_i^2$ over $p_1<p_2<q<r \equiv 1 \pmod{12}$ with $p_2^2 \le 24p_1^3+1$ and $q, r \in (p_2, 2P)$ gives an expected count $\approx 6.3\times10^{-7}$ (`python3 -m agrawal.k4_heuristic`), about 96% of it from $p_1 = 13$ (where $n < 1.1\times10^{11}$, long excluded). The mass left after Theorem B ($p_1 > 5\times10^5$) decays like X^{-4} and is of order 10^{-25}.

A proof that k = 4 is impossible would need a bound on p_1 itself. By Theorem A, fixing p_1 leaves a finite search over (p_2, q, r), all bounded polynomially in p_1. The identity $d\cdot e\cdot s\cdot \Pi = 8p_1^3$ with $p_1 \nmid des$ makes $\delta = des - 8p_1^3$ a nonzero integer ($\delta \ge 1$ if $\Pi < 1$, $\delta \le -1$ if $\Pi > 1$) equal to $8p_1^3(1/\Pi - 1)$; this ties the sizes of p_2, q, r to p_1 but does not by itself exclude them. Natural next steps: analyse δ for small values; a 128-bit version of the enumerator would reach $B = 10^{32}$ or $p_1 \le 10^6$ in about half a day each.